MIL-T-38760
1 APRIL 1966
Superseding
MIL-T-4135(USAF)
5 July 1950
MIL-T-7638(USAF)
6 June 1952
MIL-T-7639(USAF)
6 June 1952

MILITARY SPECIFICATION

TENSIOMETER DIAL INDICATING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers clamp mounting cable tensiometers.

1.2 Classification. The tensiometers shall be of the following types and ranges.

TYPE	DIAL RANGE
C7	300 lbs. to 1600 lbs.
C8	10 lbs. to 200 lbs.
C9	150 lbs. to 450 lbs.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes Wood, Nailed and Lock-Corner
PPP-B-636	Box Fiberboard

FSC 6635

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Military

MIL-P-116	Preservation Methods of
MIL-G-822	Glove Shells, Leather, M-1949
MIL-G-835	Glove Inserts, Wool, M-1949
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-W-1511	Wire Rope, Steel (Carbon) Flexible Preformed
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Specifications and standards for all materials, processes, parts and equipment which are not specifically designated herein, and which are necessary for the execution of this specification shall be selected in accordance with MIL-STD-143.

3.1.1 Standard Parts. AN standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they develope suitable properties and are replaceable by the AN standard parts without alteration, and provided the corresponding AN part numbers are referenced on the drawings and in the parts list. In applications for which no suitable corresponding AN part is in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2 Design. The tensiometer shall be designed to operate by clamping to the cable with a lever action, or equivalent fast acting device suitable for the purpose. The design shall permit operation by a mechanic using only one hand while wearing winter gloves. Finger holes, if used, shall be large enough to accommodate the use of large size mechanics' winter gloves, in accordance with MIL-G-822 and MIL-G-835.

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3.3 Construction. The tensiometer shall be designed and ruggedly constructed for long life.

3.4 Performance. The tensiometer shall be capable of operating in the temperature range of -65°F to +160°F.

3.5 Capacity.

3.5.1 Type C7. The tensiometer shall be suitable for measuring the tension in 3/16, 7/32, 1/4 and 5/16 inch diameter aircraft control cables over a range of tensions from 300 to 1600 pounds.

3.5.2 Type C8. The tensiometer shall be suitable for measuring the tension in 1/16, 3/32, 1/8, 5/32, 3/16, 7/32 and 1/4 inch diameter aircraft control cables over a range of tensions from 10 to 200 pounds.

3.5.3 Type C9. The tensiometer shall be suitable for measuring the tension in 1/8, 5/32, 3/16, 7/32 and 1/4 inch diameter aircraft control cables over a range of tension from 150 to 450 pounds.

3.6 Accuracy.

3.6.1 Type C7. The tensiometer shall be calibrated to an accuracy of $\pm$5 percent at increments of load up to and including 550 pounds and 2-1/2 percent at all increments of load over 550 pounds for each size of cable represented on the scale. The difference between successive check tests at any cable tension shall not exceed 3 percent.

3.6.2 Type C8. The tensiometer shall be calibrated to an accuracy of 5 percent over the range of 10 to 200 pounds for the 1/16, 3/32, and 1/8 inch diameter cable sizes, over the range of 30 to 200 pounds for the 5/32 and 3/16 inch diameter cable sizes, and over the range of 80 to 200 pounds for the 7/32 and 1/4 inch diameter cable sizes. The difference between successive check tests at any cable tension shall not exceed 3 percent.

3.6.3 Type C9. The tensiometer shall be calibrated to an accuracy of 5 percent over the range of 150 to 450 pounds for the 1/8, 5/32, 3/16, 7/32 and 1/4 inch diameter cable sizes. The difference between successive check tests at any cable tension shall not exceed 3 percent.

3.7 Direct Reading Dial. The Type C8 and C9 tensiometers shall have a direct reading dial calibrated in pounds of tension. A means shall be provided for locking the dial needle at any position produced by the tension load on the cable. This will permit the removal of the tensiometer from the cable for reading.

3.7.1 The deflection of the dial needle under full load shall be steady and indicate uniform stressing.

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3.7.2 There shall be no stop at the zero reading of the tensiometer. A means shall be provided for adjusting the needle to the zero reading to compensate for wear and adjustment.

3.8 Adapters. On the Type C7 tensiometer, interchangeable cable adapters, if provided, shall be held securely in place on the tensiometer, and in the carrying case by spring clips or equivalent means. The C8 and C9 tensiometers shall be self contained units requiring no adapters.

3.9 Cable Diameter Gage. The Type C8 and C9 tensiometers shall have incorporated as an integral part, a gage, for measuring unknown cable diameters.

3.10 Calibrated Test Bar. On Type C8 and C9 tensiometers, a calibrated test bar shall be supplied to provide a means of checking the calibration of the instrument. This test bar shall be a ground steel bar carefully calibrated to read the load marked on the bar when the tensiometer is preset to 1/16 inch for Type C8 and to 1/8 inch for Type C9.

3.11 Interchangeability. All parts having the same manufacturers' part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturers part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.12 Dimensions. The dimensions of the type C8 and C9 shall not exceed those shown on Figure 1. The type C7 tensiometer shall have overall dimensions not to exceed 6-1/2 x 4-1/2 x 1-1/2 inches.

3.13 Weight. The weight of the tensiometer shall not exceed three pounds.

3.14 Carrying Case. Each tensiometer shall be provided with a carrying case substantially constructed of wood, plastic or metal. The carrying case shall be designed for permanence and durability. The tensiometer shall fit into the case snugly and shall not rattle.

3.15 Workmanship. The workmanship shall be in accordance with high grade aircraft instrument manufacturing practices.

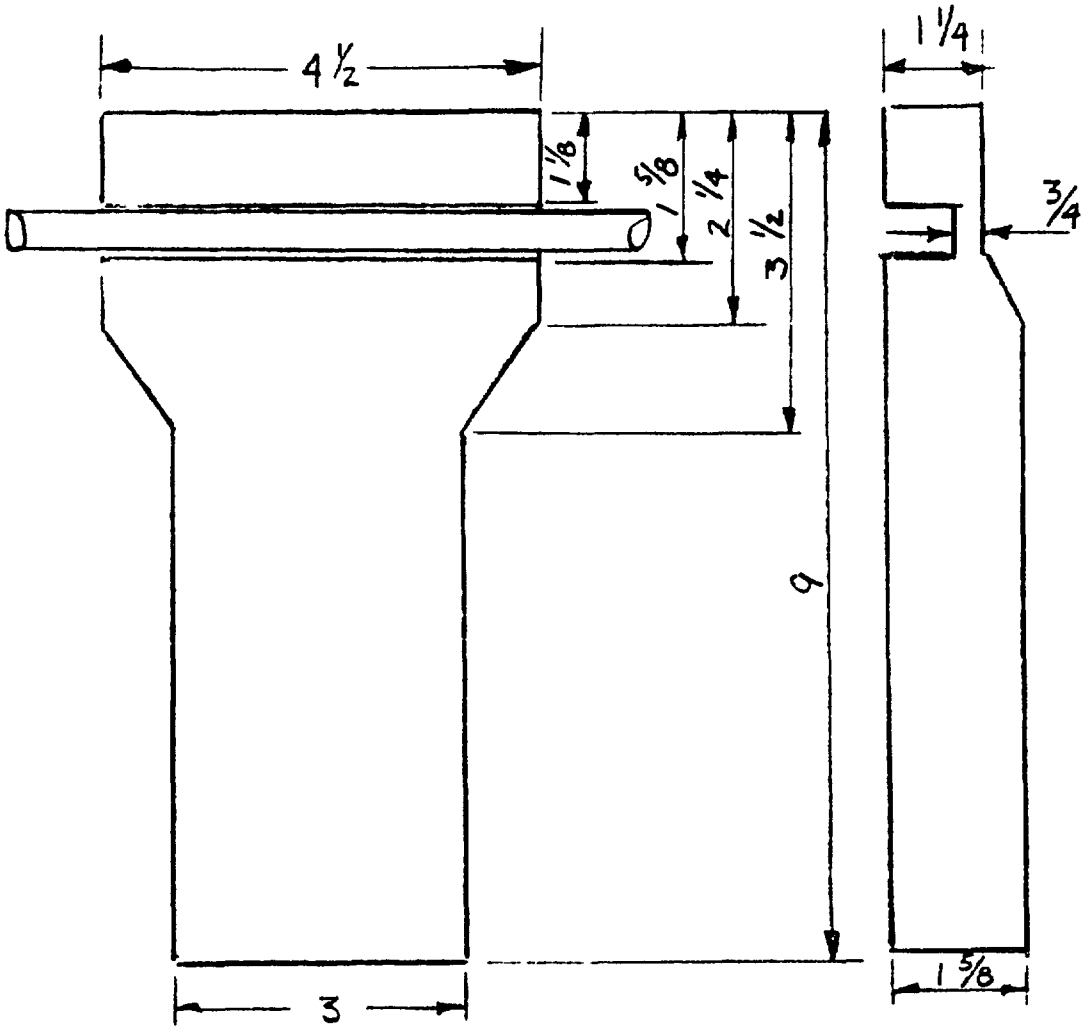
3.16 Finish.

3.16.1 Steel Parts. All steel parts shall be cadmium plated in accordance with QQ-P-416, or other suitable plating to match the finish of the tensiometer.

3.16.2 Aluminum Parts. All aluminum alloy parts may be anodized in accordance with MIL-A-8625, or polished to match the general finish of the tensiometer.

3.17 Instructions and Charts. An instruction sheet containing pertinent information on the adjustment and use of the tensiometer shall be furnished with each instrument.

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DIMENSIONS IN INCHES

FIGURE I TENSIOMETER

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3.17.1 Calibration. The type C7 tensiometer shall be provided with a calibration chart. It will include the dial readings for each increment of load and each size cable. The chart shall be attached to the inside of the cover of the carrying case and shall be protected with a transparent material.

3.18 Identification of Product. Assembly and parts shall be marked for identification in accordance with MIL-STD-130.

3.18.1 Nameplate. A nameplate, permanently and legibly filled in with the following information, shall be securely attached to the tensiometer:

TENSIOMETER
TYPE _____
SPECIFICATION NO. _____
MANUFACTURER'S PART NO. _____
MANUFACTURER'S SERIAL NO. _____
CONTRACT NO. _____
STOCK NO. _____
MANUFACTURER'S NAME OR TRADEMARK
U. S. PROPERTY

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Calibration and Check Tests. The tensiometer shall be calibrated and check tested on standard aircraft cables conforming to MIL-W-1511, by the dead weight load method described herein or any other method which gives the same results.

4.2.1 The test cable shall be approximately six feet long and shall be suspended vertically, and loaded with calibrated weights applied by a suitable loading platform.

4.2.2 The calibration loading shall progress over the full range of the tensiometer at increments of 50 pounds. The cable shall be restrained from unwinding under load by a restraining element, which will not add or detract from the loads on the cable.

4.2.3 Each tensiometer shall be calibrated for each increment of loading and for each cable size, by clamping to the cable in the manner that it will be clamped in service. At least five readings shall be taken at each load in order to determine a reliable average value.

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4.3 The calibration and check test shall be performed at -65°F and +160°F.

4.4 Tensiometers failing to meet the accuracy requirements of Section 3 when calibrated in the method set forth in 4.2 shall be rejected.

4.5 Rejected tensiometers shall not be resubmitted for inspection without furnishing full particulars concerning previous rejections and measures taken to overcome the defects.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packing.

5.1.1 Level A. The tensiometer, calibration and instruction cards, repair parts, tools and technical publications shall be contained in the carrying case. The carrying case shall be preserved in accordance with MIL-P-116, Method IC-5, with Type P-1 preservative applied to unpainted surfaces of the tensiometer and tools. The container shall be a fiber box conforming to PPP-B-636, Type I Class 2, V3C, Style RSC. The box shall be closed and sealed in accordance with the appendix to the box specification.

5.1.2 Level C. The tensiometer, calibration cards, instruction card, repair parts, tools and technical publications shall be contained in the carrying case. The carrying case shall be preserved and packaged to afford protection against deterioration and damage in accordance with the manufacturers' standard practice.

5.2 Packing.

5.2.1 Level A. The tensiometer shall be packed in nailed wood boxes conforming to PPP-B-621, Class 2 Style optional, or in wood cleated-plywood boxes conforming to PPP-B-601, overseas type style optional. The boxes shall be close-fitting and shall be closed and strapped in accordance with the appendix to the applicable box specification, except flat steel strapping shall be Type I, Class B. The gross weight of each box shall not exceed 200 pounds.

5.2.2 Level B. The tensiometer shall be packed in nailed wood boxes conforming to PPP-B-621, Class I style optional; wood cleated-plywood boxes conforming to PPP-B-601, domestic type style optional, wirebound wood boxes conforming to PPP-B-585 Class I or 2, style optional, or wood-cleated fiberboard boxes conforming to PPP-B-591 domestic type, style optional. The boxes shall be close-fitting and the gross weight of each box shall not exceed 200 pounds.

5.2.3 Level C. The tensiometer shall be packed in containers complying with Uniform Freight Classification Rules or with National Motor Freight Classification Rules or with other carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required, packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended Use. The tensiometer is intended for use in checking tension in aircraft control cables.

6.2 Ordering Data.

a. Title number and date of this specification.

b. Type number of tensiometer required.

c. Selection of applicable levels of preservation, packaging and packing.

Custodians:

Army - MO
Navy - WP
Air Force - 67

Reviewers:

Army - MO, MI
Navy - WP
Air Force - 67

User:

Army - EL

Preparing Activity:

Air Force - 67

Project No. 6635-0028

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

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